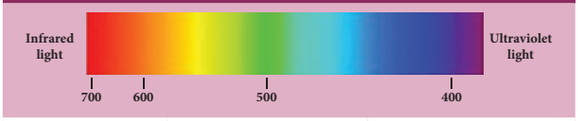
Table 5.1 visible region, frequency and wavelength of different types of radiation



Type of Radiation	Frequency Range (Hz)	Wavelength Range
gamma-rays	10^{20}-10^{24}	$<10^{-12}$m
x-rays	10^{17}-10^{20}	1 nm - 1pm
ultraviolet	10^{15}-10^{17}	400 nm - 1nm
visible	$4 - 7.5 \times 10^{14}$	750 nm - 400nm
near-infrared	1×10^{14}-4×10^{14}	2.5 μm - 750nm
infrared	10^{13}-10^{14}	25 μm - 2.5μm
microwaves	3×10^{11}-10^{13}	1 mm - 25 μm
radio waves	$< 3 \times 10^{11}$	> 1 mm

X-rays

It is produced when there is a sudden deceleration of high speed electrons at high-atomic number target, and also by electronic transitions among the innermost orbits of atoms. The wavelength range 10^{-13} m to 10^{-8} m and frequency range are 3×10^{21} Hz to 1×10^{16} Hz. X-rays have more penetrating power than ultraviolet radiation. X-rays are used extensively in studying structures of inner atomic electron shells and crystal structures. It is used in detecting fractures, diseased organs, formation of bones and stones, observing the progress of healing bones. Further, in a finished metal product, it is used to detect faults, cracks, flaws and holes.

Gamma rays

It is produced by transitions of atomic nuclei and decay of certain elementary particles. They produce chemical reactions on photographic plates, fluorescence, ionisation, diffraction. The wavelength range is 1×10^{-14} m to 1×10^{-10} m and frequency range are 3×10^{22} Hz to 3×10^{18} Hz. Gamma rays have high penetrating power than X-rays and ultraviolet radiations; it has no charge but harmful to human body. Gamma rays provide information about the structure of atomic nuclei. It is used in radio therapy for the treatment of cancer and tumour, in food industry to kill pathogenic microorganism.

EXAMPLE 5.4

A magnetron in a microwave oven emits electromagnetic waves (em waves) with frequency f = 2450 MHz. What magnetic field strength is required for electrons to move in circular paths with this frequency?.

Solution

Frequency of the electromagnetic waves given is f = 2450 MHz

The corresponding angular frequency is

$$\omega = 2\pi f = 2 \text{ x } 3.14 \text{ x } 2450 \text{ x } 10^6$$
$$= 15{,}386 \text{ x } 10^6 \text{ Hz}$$
$$= 1.54 \times 10^{10} \text{ s}^{-1}$$

The magnetic field $B = \dfrac{m_e \omega}{|q|}$

Mass of the electron, $m_e = 9.22 \text{ x } 10^{-31}$ kg

Charge of the electron

$$q = -1.60 \times 10^{-19} C \Rightarrow |q| = 1.60 \times 10^{-19} C$$

$$B = \frac{\left(9.22 \times 10^{-31}\right)\left(1.54 \times 10^{10}\right)}{\left(1.60 \times 10^{-19}\right)} = 8.87425 \times 10^{-2} T$$

B = 0.0887 T

This magnetic field can be easily produced with a permanent magnet. So, electromagnetic waves of frequency 2450 MHz can be used for heating and cooking food because they are strongly absorbed by water molecules.

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